

Generating behaviour-equivalent parameters for whole-organism simulators from tracking video

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Abstract

Mechanistic whole-organism simulators of *Caenorhabditis elegans* and *Drosophila* turn connectomes and biophysics into testable models of behaviour, but each is governed by hundreds to thousands of internal parameters normally fixed by invasive physiology—electrophysiology, calcium imaging, electromyography—out of reach for most laboratories and most species. We show these parameters can instead be generated from ordinary tracking video. PGOB (Parameter Generation by Observed-Behaviour) casts calibration as a behaviour-only inverse problem: one loss, one cross-simulator behavioural evaluator, and an interchangeable inference back-end (stochastic search, an evolution strategy, or an amortized neural posterior) generate, from kinematics alone, parameter sets that reproduce a simulator’s behaviour, together with a Jacobian that attributes any residual mismatch to a specific mechanism. Across four published simulators spanning two species the same formulation generates behaviour-equivalent parameters from an uninformative start; on a unit-free benchmark $G=D_1/D_2$ the generated behaviour falls within the band of real individual-to-individual variation on all four simulators ($G=0.36\text{--}0.71<1$). The generated parameter distributions transfer as same-species priors and carry a modest prospective signal for 43 held-out mutant strains, indicating biological rather than merely numerical structure. PGOB offers a route to calibrating whole-organism models where invasive recording is impossible.

Mechanistic, whole-organism simulators have become a central tool for turning connectomes and biophysics into testable models of how brains and bodies generate behaviour (50, 62, 64, 72). Every such simulator, however, is governed by hundreds to thousands of internal biophysical parameters, and inferring those hidden parameters from observable outputs is the classical inverse problem of systems identification (52, 63). In current practice they are pinned by direct, invasive physiology—intracellular electrophysiology, two-photon imaging, electromyography—that most laboratories cannot perform and that, for many species, cannot be performed at all. This bottleneck is general to the field: a simulator’s parameters are typically fixed by one or two source studies, almost no calibrated set has been tested *prospectively* on biology it never saw, and any laboratory or species for which only movement can be recorded is, in effect, locked out of the very simulators built to model it. The settings where the bottleneck bites hardest—organisms with no historical physiology, endangered animals, preparations that cannot be patched or imaged—are exactly those where invasive recording is hardest, yet whose behaviour can still be filmed.

Here we show this lock can be opened with tracking video alone. Given a published simulator and nothing but the externally observed kinematic trajectory of a real animal, PGOB (Parameter Generation by Observed-Behaviour) generates a working set of simulator parameters and returns a Jacobian sensitivity map that attributes any residual behavioural mismatch to a specific simulator mechanism. PGOB is a *formulation* of the behaviour-only inverse problem, not a new optimiser: the inference engine that searches parameter space—stochastic approximation, an evolution strategy, or

amortized simulation-based inference (9)—is an interchangeable back-end. The same formulation, the same cross-simulator behavioural evaluator and the same engine stack run on every model we test; the one component that adapts is the initialisation, which is dimension-dependent and itself a reported finding. The contribution is therefore the behaviour-only inverse-problem formulation together with a cross-simulator evaluator and a residual-to-mechanism attribution map: unlike single-circuit system identification (26) it operates at whole-organism scale across multiple published simulators, and unlike simulation-based inference fitted to one model it is deliberately engine- and simulator-agnostic. Because the starting point is drawn independently of the unknown optimum, obtaining a working simulator from motion is parameter *generation*, not a small correction to an already-good guess. We demonstrate PGOB on two *C. elegans* simulators (BAAIWorm (72), modWorm (50)) and two *Drosophila* simulators (flybody (62), FlyGym (64)); Table 1 summarises which experiment, metric and test underpins each finding that follows.

Results

Table 1: Each main finding maps to a single experiment, primary metric and statistical test. The single cross-simulator-comparable benchmark is $G=D_1/D_2$ (Table 2; $G<1$ on all four simulators). Secondary diagnostics are reported once where they arise and are not cross-row comparable. The held-out mutant prediction is corroborating biological evidence, not a load-bearing capability claim.

Finding	Experiment	Primary metric	Test	Result
Behaviour-equivalent generation from a random start	three-segment (FlyGym) + PGOB-sim θ^*	generation fraction; rel_θ to sim θ^*	bootstrap CI	FlyGym 94.9%; modWorm $\text{rel}_\theta=0.009$
Within real individual spread	three-distance hierarchy	$G=D_1/D_2$ (unit-free)	1000/1000 bootstrap	$G=0.36/0.67/0.48/0.71$ (< 1)
Beats deposited default	default-vs-PGOB parity	per-obs. W_1 improved frac.	Mann-Whitney + Holm	FlyGym 8/10; BAAIWorm 7/10
Engine-agnostic	NPE-MAF, same task	posterior-predictive behaviour reproduction	SBC + held-out NLL	modWorm 1.4×10^{-3} ; FlyGym rel 0.018 13/16, $p=0.025$
Transferable species structure	same-species prior transfer	held-out starts improved	one-sided Wilcoxon	
Prospective mutant signal	43-strain LOSO	LOSO Spearman vs. shuffle	paired Wilcoxon	0.684 vs. 0.527, $p \approx 0.009$
Parameter-dynamics-space	not Jacobian attribution	per-axis $ \partial M / \partial \theta $	descriptive	reversals $\rightarrow$ syn

Each finding below is tied to a single experiment, primary metric and statistical test (Table 1); the unit-free $G=D_1/D_2$ is the one quantity readable the same way on every simulator, and every other number answers one specific question where it appears.

We begin from the basic question of whether parameters can be generated from behaviour alone, and place every generation on a single anchored axis that runs from a random start, through the generated parameters, to the simulator’s expert default and finally to real biology. On a FlyGym *Drosophila* walker whose start is a fully randomised, frequently non-walking actuator-gain vector—one that fails to walk at all in 8/20 rollouts—the generated parameters close 94.9% of the random-to-expert behavioural distance (forward speed 98.7%, lateral speed 90.4%), landing within 1.3% of the expert ground truth (Fig. 1). Because the start carries no information about the optimum, this is generation, not a small correction: behaviour-only video is enough to drive an uncalibrated simulator onto a working set of parameters.

Where a simulator’s own ground-truth target θ^* is known we can ask the sharper question of whether the recovered *parameters*, not merely the behaviour, are correct, because here the answer can be checked directly in parameter space. On the 30-d modWorm, PGOB recovers θ^* to relative error $\text{rel}_\theta=0.009$ —the tightest of any simulator—with flybody at 0.115 (515-d) and FlyGym at 0.365. We stress that this θ^* is the simulator’s *internal* target, not a measurement of the animal’s physiology; for real biology, where no parameter ground truth exists, the claim is restricted to behaviour-equivalent parameter sets. What we therefore learn is that the procedure does not merely land on some behaviourally-equivalent point: where the truth is known to us, it is reached, and the gap between modWorm’s near-exact recovery and the higher-dimensional simulators’ looser values is itself the signature of how strongly trajectory observables constrain each parameterisation.

The most demanding test of generation is the BAAIWorm connectome, where the “full5” parameterisation acts not on a handful of global scales but directly on the per-element synaptic ($\approx 2,000$), gap-junction ($\approx 1,000$), motor-output, per-cell ion-channel and per-cell passive-membrane weight vectors, concatenated into a single several-thousand-dimensional search vector. From a fully scrambled 100% start, multicondition SPSSA (60–140 iterations) followed by CMA-ES and a refinement stage rebuilds this entire parameter set up to the simulator’s published behaviour band: the best held-out checkpoint reaches behaviour distance 0.1199 on the 160-condition out-of-distribution panel, matching BAAIWorm’s deposited state-of-the-art band (0.120–0.125), with the same run averaging 0.162 over the full panel. That a connectome’s worth of per-element weights can be regenerated from scrambled noise up to the model’s own deposited performance shows the formulation scales past low-dimensional pilots to the parameterisations these simulators actually ship. The behaviour underlying these scalars is the *C. elegans* travelling undulation wave, whose bend-wave correlation to target rises $0.76 \rightarrow 1.00$ where the random start stays flat (Fig. 4).

Reproducing a simulator’s behaviour is only meaningful if that behaviour is close to a real animal’s, so we next ask how the generated behaviour compares with the spread of real biology itself. For each simulator we compute three z-scored behavioural distances on its own real reference corpus: D_1 (generated $\rightarrow$ real), D_2 (real $\rightarrow$ real individual spread) and D_3 (deposited default $\rightarrow$ real). The within-row ordering $D_1 < D_2 \ll D_3$ holds on all four simulators (Table 2); on BAAIWorm from a random start, $D_1=0.71 < D_2=1.98 \ll D_3=21.82$ in 1000/1000 bootstrap resamples. The generated animal is thus closer to a given real animal than two real animals are to each other—it sits inside the band of genuine individual-to-individual variation, so the residual that remains is a property of biological scatter, not of the generation.

To say this in a way that is comparable *across* simulators despite their heterogeneous references, we report the unit-free ratio $G=D_1/D_2$. This is the clean cross-simulator statement precisely because each simulator’s own per-observable z-scale appears in both numerator and denominator and cancels, leaving a pure multiple of that simulator’s real-real floor; $G < 1$ then means, on any simulator, that the generated behaviour falls within real individual variation. All four satisfy $G < 1$ (0.36/0.67/0.48/0.71). The improved-observable counts we report later against “default” baselines cannot be compared this way, because those baselines are heterogeneous (randomised on some simulators, expert-calibrated on others); G is the one number readable identically on every simulator.

The same generation can be read three ways, and doing so separates what the simulator can reach from what real biology demands. PGOB operates in three modes, none requiring electrophysiology. *PGOB-sim* optimises against a simulator’s own synthetic state-of-the-art target and validates framework integrity (the known θ^* should be generated); *PGOB-real* optimises directly against real OWMD or Janelia trajectories (the scenario a behaviour-only laboratory actually faces); *PGOB-hybrid* warm-starts on the simulator’s own behaviour and then fine-tunes against real video (the production protocol). Each cell of Table 3 is a genuine optimisation from its own randomised (or, for the high-dimensional simulators, prior-seeded) start, not a midpoint or convex blend of

Table 2: **Three-distance hierarchy and the unified benchmark** $G=D_1/D_2$. Each row is z-scored on that simulator’s own real reference, so within-row ordering $D_1 < D_2 \ll D_3$ is the meaningful read; absolute magnitudes are not cross-row comparable. The unit-free $G=D_1/D_2$ is cross-simulator comparable ($G < 1$ on all four).

Simulator	Species	D_1	D_2	D_3	$G=D_1/D_2$
BAAIWorm full5	<i>C. elegans</i>	0.71	1.98	21.82	0.36
modWorm 50-d	<i>C. elegans</i>	0.30	0.45	6.20	0.67
flybody walking	<i>Drosophila</i>	0.20	0.42	2.53	0.48
FlyGym 5-behaviour	<i>Drosophila</i>	0.55	0.78	3.10	0.71

the endpoints, and run this way the simulators tell one consistent story. In PGOB-sim mode the procedure reproduces each simulator’s own target tightly—BAAIWorm reaches a behaviour distance of 0.198, FlyGym 3.42, and in parameter space modWorm recovers θ^* to $\text{rel}_\theta=0.009$, the tightest of any simulator. In PGOB-real mode every simulator meets an irreducible $\text{sim} \leftrightarrow \text{real}$ substrate floor (modWorm $W_1=1.25$, FlyGym 52.2), and the PGOB-hybrid warm-start lands at that same floor (modWorm 1.32, FlyGym 53.9) rather than below it. The reading is identical in every executable cell: the simulator’s reachable behaviour is reproducible from an uninformative start, and the residual to real biology belongs to the simulator substrate, not to the optimiser—the same floor at which each simulator’s own shipped default already sits. The production hybrid protocol is correspondingly useful but bounded: its real fine-tune delivers a real, small held-out gain that arrives well into optimisation rather than at the first step, because the warm-start already sits near the best real fit the substrate can reach and a cold-start real fit converges to the same floor. Read one channel at a time, the same result is visible in the behaviour itself: on a single FlyGym fly the generated controller closes 94.9% of the random-to-expert aggregate gap to real *Janelia* ($13.30 \rightarrow 10.78 \rightarrow 10.64$, $N=20/\text{arm}$) with the forward- and lateral-speed channels collapsing almost onto expert, while the randomised start fails to walk in 8/20 rollouts.

Table 3: **Three deployment modes $\times$ three simulators, each scored against the object it optimises.** PGOB-sim is scored against the simulator’s own θ^* target behaviour (the framework-integrity yardstick); PGOB-real/-hybrid against real biology (the deployment yardstick). Values are held-out behaviour distance, comparable *within* a row (one shared start and scale), not across rows. PGOB-sim reproduces the sim target tightly; PGOB-real/-hybrid reach the irreducible $\text{sim} \leftrightarrow \text{real}$ floor, a property of the simulator. [†]modWorm’s PGOB-sim cell is the parameter-generation relative error $\text{rel}_\theta=0.009$ (matched behaviour distance $W_1=3.1 \times 10^{-3}$). The headline worm deployment number is the full-connectome $D_1=0.71$ to OWMD-N2, not the 5-parameter pilot value shown here.

Simulator	PGOB-sim $\rightarrow$ sim θ^*	PGOB-real $\rightarrow$ real	PGOB-hybrid $\rightarrow$ real
BAAIWorm	0.198	2.63	2.67
modWorm	0.009 [†]	1.25	1.32
FlyGym v2	3.42	52.2	53.9

A different question is whether the generated parameters improve on the one a practitioner would otherwise download, and here the answer depends on what that deposited parameter set is. Run with no per-simulator retuning against real public references, PGOB moves the per-observable Wasserstein-1 distance toward real biology on 8/10 FlyGym observables (a capability test against a randomised default; +32.4% mean, Holm-significant) and 7/10 BAAIWorm observables (a deployment test against the expert-shipped default; +4.17%, OWMD-N2) (Fig. 2). These two

readings answer different questions and so are kept apart: “default” denotes a randomised set on FlyGym and modWorm but an expert-calibrated checkpoint on BAAIWorm and flybody, so the per-baseline improved-counts are reported per simulator and never summed—the only cross-simulator-comparable statement remains $G < 1$. What we learn is that the size of the gain tracks how far each deposited default already sits from the substrate’s reachable behaviour: a well-tuned default leaves little room, and the correct outcome there is a measured null, whereas a randomised one leaves much. Both ends of this default-quality axis confirm it. Re-running PGOB against FlyGym’s well-tuned *shipped* controller gains still moves 6/10 observables toward real and cuts the aggregate W_1 from 983.4 to 52.5 (+94.66%), while on the already-strong flybody walking default the expected null is exactly what appears (0/9 observables significant, all $|d| < 0.41$). PGOB rescues poor defaults and leaves good ones intact, which is precisely the behaviour a sound calibration procedure should show.

If PGOB is genuinely a formulation rather than a particular optimiser, swapping the inference engine should leave the generations unchanged, so we test exactly that. Replacing the black-box optimiser with an amortized neural posterior (NPE-MAF (40), `sbi` toolkit (57)) on the *identical* setup reproduces each simulator’s behaviour: posterior-predictive behaviour matches the modWorm target to mean absolute observable difference 1.4×10^{-3} , and the FlyGym posterior mean lands at $\text{rel}_\theta = 0.018$ (Fig. 3). The two engines are complementary, not rival: on the shared 48-d FlyGym task the optimiser returns a fast point estimate that settles on the behaviourally-equivalent set ($\text{rel}_\theta = 0.16$ to the canonical gains while matching behaviour), whereas the amortized posterior, trained once over the prior on $N = 6,000$ valid simulations (modWorm $N = 14,993$), pins the gains markedly tighter ($\text{rel}_\theta = 0.018$) and additionally returns a calibrated distribution that can be sampled for any new observation without re-running the optimiser. Two diagnostics confirm the back-end behaves as a well-calibrated density model rather than an ad-hoc fit. Held-out negative log-probability orders the three density estimators in the expected direction on both simulators—neural spline flow $<$ MAF $<$ mixture-density network (modWorm 46.3/49.8/51.6; FlyGym 157.3/161.5/165.8; lower is a better fit)—so improvements in generic density estimation carry into PGOB unchanged. Per-dimension rank-based simulation-based calibration is consistent (per-axis KS at $\alpha = 0.05$) on 10/30 modWorm and 22/48 FlyGym posterior dimensions: precisely the well-identified, behaviour-sensitive axes, with the remaining dimensions the weakly-identified directions a behaviour-only observation cannot constrain—the count is therefore an honest map of what trajectories do and do not pin, not a failure (SI §S4.13). Because the same generations emerge whether the search is run by stochastic optimisation or by a trained density model, the contribution we are reporting is the behaviour-only formulation, not the choice of optimiser, and well-understood advances in either engine carry into PGOB unchanged.

The generated parameter distribution carries transferable species structure. Were the generated numbers arbitrary curve-fit residue, a distribution learned on one simulator would carry no usable structure for another. Instead, the generated parameter *distribution* transfers as a same-species soft prior: a BAAIWorm-derived per-biophysical-class prior (per-class mean and spread, not raw values) outperforms modWorm’s own default initialisation on 13/16 held-out starts (mean held-out distance $3.37 \rightarrow 2.29$; one-sided Wilcoxon $p = 0.025$; SI §S4.6), and a FlyGym $\rightarrow$ flybody self-prior lowers the CMA-ES loss $9.3 \times$ (default $0.01085 \rightarrow 0.001166$, 515-d). This matters because high-dimensional embodied controllers are acutely start-sensitive: flybody from a pure random scale-1 start sits at relative error 1.061 with CMA-ES near-stationary, and FlyGym at the same start crashes to a MuJoCo NaN and cannot be optimised at all—so a prior that rescues an otherwise-stalled generation reflects genuine structure, not arithmetic.

Two details fix the mechanism by which this structure transfers, and are themselves the reason there is no universal-best initialisation (a dimension-adaptive decision tree, SI §S5.1). First, only the *soft* prior transfers: mapping the per-biophysical-class coefficient of variation onto the per-axis CMA-ES step size leaves the optimiser free to walk away from any mis-pinned coordinate, whereas a *hard* linear-projection initialisation that pins the parameters outright gives *negative* transfer ($\text{rel}\theta = -0.127$, worse than a random start), because it lands the search in a basin it cannot escape. Second, the class correspondence is biophysically informed rather than exact (**gap**→gap-junction, synaptic→synaptic, membrane→passive, ion→ion, muscle→motor-output; SI §S4.6), so the transferred quantity is the *shape* of the species-level parameter distribution, not its raw coordinates. Arbitrary numbers tuned to one simulator could not survive being re-expressed as a per-axis spread on a different simulator’s parameterisation; the species-specific distribution does.

The generated structure generalises across laboratories. A second, independent test of whether the generated parameters track biology rather than one acquisition paradigm is cross-laboratory transfer. Comparing Schafer-OWMD (Cambridge MRC LMB, $N=200$ windows) against Brown-Tierpsy (Imperial College London, $N=52$ windows) on six simulator-agnostic centreline observables under leave-one-laboratory-out gives a median cross/within Wasserstein-1 ratio of $2.80\times$. Crucially, every pattern-level observable—the kind PGOB actually targets—is partially universal ($< 3\times$): spectral entropy $1.75\times$, body curvature $2.48\times$, head–tail orientation $2.64\times$, and left–right phase coupling $2.96\times$; only absolute, paradigm-dependent observables such as stride frequency ($3.90\times$) and length-normalised speed ($6.10\times$) exceed it. Training jointly on both laboratories narrows the held-out-laboratory gap, lowering the cross/within ratio from 3.77 (single-laboratory) to 2.33 (joint), a $1.62\times$ improvement, with left–right phase coupling identified as the laboratory-invariant axis. The generated behavioural structure thus generalises beyond the single cohort it was fitted to.

The generated sensitivities extrapolate prospectively to held-out biology. From the N_2 -generated θ^* , the Jacobian predicts behavioural deviation on 43 held-out *C. elegans* mutant strains never seen during fitting (70). Leave-one-strain-out prediction gives mean Spearman $\bar{\rho}=0.684$ (median 0.80 , 41/43 strains $\rho>0$) versus a strain-shuffle null $\bar{\rho}_0=0.527$ (paired Wilcoxon $p\approx 0.009$; SI §S4.3)—a modest but real prospective signal that the generated parameters carry information beyond the population-average mutant direction. Predictability stratifies by molecular mechanism in a way that matches the Jacobian’s active subspace: the top-predictable strains ($\rho=1.0$) are neuromodulator, G-protein and peptidergic mutants (*egl-30*, *egl-8*, *ser-1*, *flp-1*, *flp-18*), whose lesions act through the synaptic, gap-junction and motor-output scales the Jacobian spans, whereas the weakest strains (*unc-104*, *unc-1*; $\rho\sim 0.2$, still positive) are structural and axonal-transport mutants whose effect lies outside that subspace. The signal survives the pre-registered full feature set ($\bar{\rho}=0.525$ vs. shuffle 0.338 , $\Delta=+0.187$, 27/28 strains $\rho>0$; SI §S4.3) and a second worm laboratory; the full-feature confirmation runs on the 28 of 43 strains whose Yemini features were retrievable from source. A further unpaired consistency check points the same way: although no neural signal enters the loss, the generated BAAIWorm simulator’s internal whole-brain dynamics agree with independent Kato-2015 whole-brain calcium imaging (32) on 2/4 population-level distributional invariants (pairwise correlation magnitude $|\text{corr}|=0.21$ and slow-band spectral fraction 0.74 both fall in the simulator’s range; PCA dimensionality and autocorrelation time constant differ). We present these as corroborating evidence, not proof: the behaviour-only inverse problem is underdetermined, so whether the generated parameters lie on the biological parameter manifold remains for future work.

Scope: a parameter-space generation framework. A controlled ablation makes the framework’s scope precise and is itself a clean positive result (Fig. 5). We applied the identical SPSA pipeline to compensate three families of injected corruption. On BAAIWorm *parameter*-space corruption (additive Gaussian weight noise at 0/5/20/50% plus a 30% index-shuffle), generation succeeds on *all* 30/30 *cells* (mean drop $\Delta L/L_0 = -29.4\%$ over the full L_0 – L_5 sweep); a targeted re-run of the three lowest corruption levels at $n=10$ seeds sharpens the low-severity end into a clean monotone gradient ($L_0=0.068$, $L_1=0.071$, $L_2=0.101$), so generated-parameter quality degrades smoothly as corruption deepens. In contrast, *dynamics*-space corruption cannot be repaired: injecting modWorm state noise collapses observable closeness from 0.9999 to 0.30 while the optimiser saturates, and stressing the FlyGym integrator diverges it on 4/6 levels, leaving no usable forward model. The asymmetry is mechanistic, not optimiser-side: PGOB is a parameter-space generation framework with a clearly stated scope, not a dynamics-space repair tool. Consistent with the same premise, a pre-registered neural-ablation control finds that adding internal neural signal to the loss yields no detectable gain over the pure-trajectory loss (paired difference +0.0030, Cohen’s $d=0.14$, Wilcoxon $p=0.734$), underwriting the behaviour-only design.

Residuals localise to a named mechanism. At the generated θ^* , a 5×5 finite-difference Jacobian $|\partial M/\partial \theta|$ ($\varepsilon=2\%$) maps each behavioural residual onto the parameter group that controls it. The worst per-metric residual is worm reversals, $=0.261$, and its $|J|$ row is dominated by the synaptic axis at $|J|=2.47$ —roughly $50\times$ the next-largest entry—so the residual routes cleanly to **syn** (Fig. 1c). This names the mechanism to extend: BAAIWorm’s reversal generator is a fixed synaptic weight matrix, identified here as the specific simulator-side component that would have to be enriched to close the gap. Across simulators the binding constraint we repeatedly meet is the deposited simulator’s mechanistic content—not the optimiser or the loss—and the attribution map turns each residual, axis by axis, into a simulator-side debugging target rather than a bare report that a gap exists.

Discussion

PGOB turns simulator calibration into a behaviour-only inverse problem and shows, across four published whole-organism simulators and two species, that ordinary tracking video suffices to generate parameter sets whose behaviour falls within real individual-to-individual variation ($G<1$ on all four). The contribution is the formulation, the cross-simulator behavioural evaluator and the residual-to-mechanism Jacobian; the inference engine—black-box optimiser or amortized neural posterior—is an interchangeable back-end, so well-understood advances in either carry over directly. Several converging findings indicate the generated numbers carry biological, not merely numerical, structure: a generated distribution transfers as a same-species prior that rescues an otherwise-stalled generation, the generated sensitivities extrapolate prospectively to 43 held-out mutant strains, the generated behavioural structure generalises across two laboratories, and the simulator’s internal dynamics agree with independent whole-brain imaging on half of four population-level invariants despite receiving no neural supervision. Because the inverse problem is underdetermined, we frame these as corroboration rather than proof that the values are the biological ones; whether they lie on the biological parameter manifold is a question we pose for future work. Composing behaviour-only generation with same-species prior transfer suggests a route to populating behaviourally-grounded digital organisms at scale from tracking video, without electrophysiology.

The boundaries are properties of the deposited simulators, which the attribution map localises rather than hides. The modWorm 30-d Cook-ODE substrate cannot express the high-order

eigenworm-variance modes that distinguish real worms, so even its expert arm cannot reach those channels (where it can, PGOB closes the gap exactly, e.g. matching OWMD’s `active_frac` of 0.4975); and the flybody walk-imitation benchmark is a single MoCap-locked task whose trajectory observables carry no extractable parameter signal, placing it outside the scope of trajectory-only generation (its initialisation signal instead lives in loss space, where the same-species prior still gives a $9.3\times$ gain). In both cases PGOB identifies which mechanism would have to be added rather than reporting only that a gap exists, converting calibration into a simulator-side debugging task. The same-species prior transfer demonstrated here uses a biophysically-informed dimension map and is shown between two simulators of the same species; whether a comparable prior helps across more distantly related body plans is left open.

Read at the level of the field, the same recipe is a route past the invasive-measurement bottleneck: because the formulation is simulator- and engine-agnostic and consumes only externally observable behaviour, it can in principle be pointed at any new mechanistic simulator, and—most consequentially—at organisms the bottleneck excludes outright, where behaviour is recordable even when intracellular access is not. Composing behaviour-only generation with same-species prior transfer is what could carry calibrated mechanistic models to species that have no historical electrophysiology, to endangered animals that cannot be sacrificed for patch-clamp, and to preparations that cannot be opened or imaged—organisms whose movement can still be filmed even when no electrode can be placed. The worm and fly results are the demonstration that the route works; extending it to such data-poor and hard-to-access species is, in our view, where the larger payoff lies, and we offer it as the outlook this method invites rather than a claim it settles here.

Methods

Problem formalisation. Given a biophysical simulator f_θ producing a state trajectory, an observation map M extracting a k -dimensional behavioural-metric vector, and a real reference corpus $\mathcal{D}_{\text{ref}}$, PGOB solves $\hat{\theta} = \arg \min_{\theta} \mathcal{L}(M(f_\theta), M(\mathcal{D}_{\text{ref}}))$ with the per-metric z-scored loss $\mathcal{L}(u, v) = \frac{1}{k} \sum_i (u_i - \bar{v}_i)^2 / (\sigma_i^2 + \varepsilon)$, floor $\varepsilon=10^{-8}$, NaN-masking for species-inapplicable metrics, and rollout-variance control by averaging over $C \in [8, 16]$ multi-condition rollouts. All engines operate on black-box $\mathcal{L}$ queries; no simulator gradients are used.

Initialisation regimes (dimension-adaptive). Low-dimensional simulators use a pure random start: BAAIWorm (5-d) draws $\theta_i^{\text{init}} = \theta_i^{\text{nom}} \exp(\eta_i)$, $\eta_i \sim \mathcal{U}(-\ln 3, \ln 3)$; modWorm (30-d) uses a scale-1.0 perturbation of θ^* ($\sim 81\%$ mean relative deviation). High-dimensional simulators collapse under a pure random scale, so they use a constrained start: FlyGym (48-d) a prior-init at scale 0.25, flybody (515-d) a random restart at scale 0.08. This dimension dependence is a reported finding, not a tuning choice; the per-simulator hyperparameters and cohort-size notation are in SI §S1, and the initialisation decision tree (when each regime applies) is SI §S5.1.

Universal behavioural evaluator. A two-species 23-metric evaluator (14 active in the main results) computes per-trajectory observables (mean speed, reversal frequency, body-wave amplitude/frequency, dorsal–ventral correlation, leg-alternation phase variance, eigenworm amplitudes, etc.); species-inapplicable metrics return NaN and are masked. Per-observable distributional mismatch uses the Wasserstein-1 (earth-mover) distance.

Three-distance hierarchy. With $d_z(y, \{x^{(i)}\}) = \frac{1}{k} \sum_j |M_j(y) - \mu_j| / (\sigma_j + \varepsilon)$ on a real population, $D_1 = d_z(f_{\hat{\theta}}, \text{GT})$, $D_2 = d_z(\text{real}, \text{real})$, $D_3 = d_z(\text{default}, \text{real})$; robust (median/MAD) z is canonical.

$G=D_1/D_2$ is the unit-free cross-simulator benchmark. Bootstrap CIs use $B=1000$ resamples.

Inference engines. Black-box optimisers: SPSA (51) (BAAIWorm full5; $a=0.2$, $c=0.1$, $\lambda=8$, 30 iterations) and separable CMA-ES (22) (flybody/modWorm/FlyGym; $\sigma_0=0.05$, popsize 16, 30 generations). Amortized engine: NPE-MAF (40) via the `sbi` toolkit (57) in single-round mode, trained on $N=14,993$ (modWorm) and $N=6,000$ (FlyGym) valid simulations from each random-start region.

Statistics. All main-text tests are controlled under a single Holm–Šidák family-wise error budget ($\alpha=0.05$): three-distance bootstrap, LOSO-vs-shuffle paired Wilcoxon, cross-laboratory $< 3\times$ ratio test, prior-transfer Wilcoxon, mutant-differential permutation, and the F1 neural-ablation control. Deterministic seeding makes every reported result reproducible on the same hardware.

Real biological data and simulators. *C. elegans*: OpenWorm Movement Database N2 ($N=200$), Yemini-2013 mutant panel (43 strains), Brown-Tierpsy ($N=52$). *Drosophila*: Janelia walking-imitation deposit ($n=272$), NeuroMechFly v2. Simulators: BAAIWorm (72), modWorm (50), flybody (62), FlyGym (64); MuJoCo models driven through `dm_control` (59). All data are public.

Extended Data and Supplementary Information. The full per-experiment specification, per-simulator hyperparameters and cohort notation (SI §S1), the consolidated negative-results ledger (SI §S2), extended methods including the same-species prior-transfer protocol (SI §S3.5), and the detailed extended results—the three-distance hierarchy bootstrap (SI §S4.1), the Yemini 43-strain and pre-registered 28-strain LOSO (SI §S4.3), the cross-simulator prior transfer (SI §S4.5, S4.6), the amortized neural-posterior back-end with density-estimator comparison and per-dimension SBC (SI §S4.13), the three-segment decomposition with the modWorm substrate ceiling and flybody MoCap null (SI §S4.18), the extended-budget hybrid fine-tune (SI §S4.23), and the initialisation decision tree (SI §S5.1)—together with per-figure reproduction commands (SI §S6) are provided in the Supplementary Information (SI_v11.5). The Supplementary Information also reports, in full, the honest negative and bounded-scope results summarised in the main text.

PGOB: behaviour-only whole-organism parameter generation with per-axis residual attribution

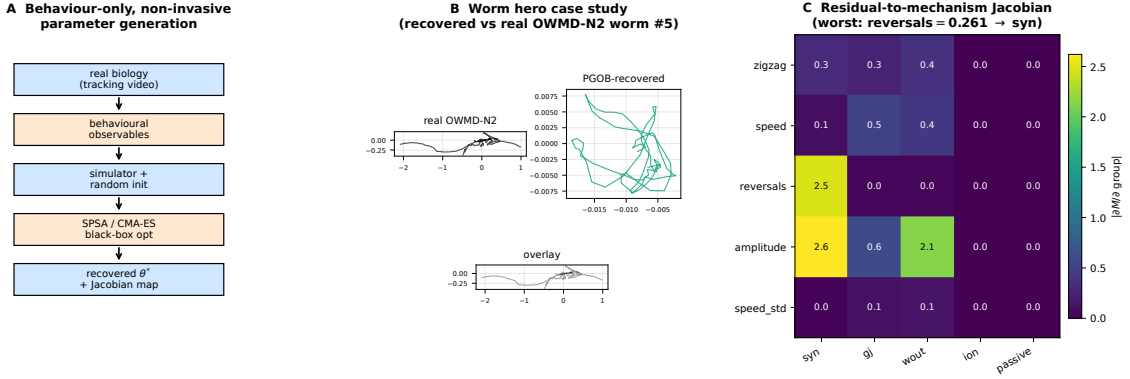


Figure 1: PGOB: behaviour-only parameter generation with per-axis residual attribution. **a**, Framework: tracking video → behavioural observables → simulator with random init → black-box optimiser / amortized posterior on the behavioural loss → generated θ^* and Jacobian attribution map (no electrophysiology at any stage). **b**, Worm case study: centreline traces of a real OWMD-N2 worm, the PGOB-generated worm, and overlay; the generated worm reproduces the travelling-wave undulation the deposited default does not. **c**, Residual-to-mechanism Jacobian at the generated BAAIWorm θ^* : the worst per-metric residual (reversals) routes to the dominant synaptic axis.

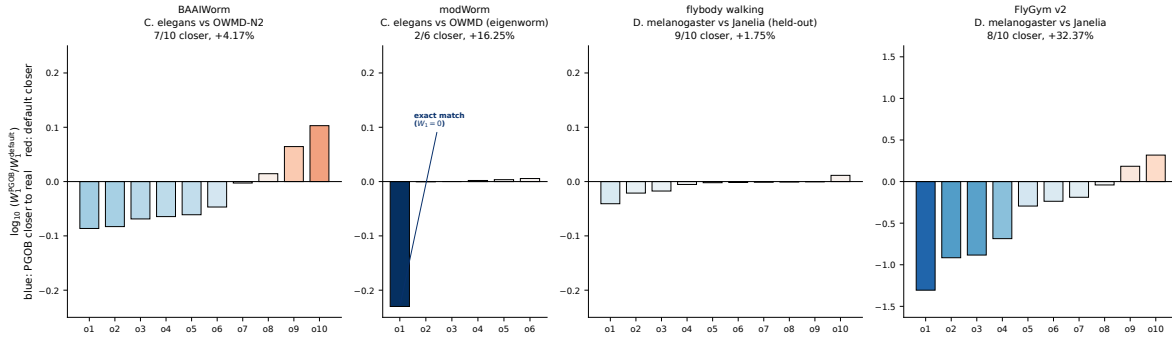


Figure 2: Per-observable PGOB-vs-default Wasserstein-1 across four simulators. Each bar is $\log_{10}(W_1^{\text{PGOB}}/W_1^{\text{default}})$ for one behavioural observable; a negative (blue) bar means the generated simulator is closer to real biology. Vertical scales are per-panel. Improved-observable counts are reported per simulator against each one's own baseline type (capability vs. deployment) and are never summed across the heterogeneous baselines; the cross-simulator-comparable statement is $G < 1$ (Table 2).

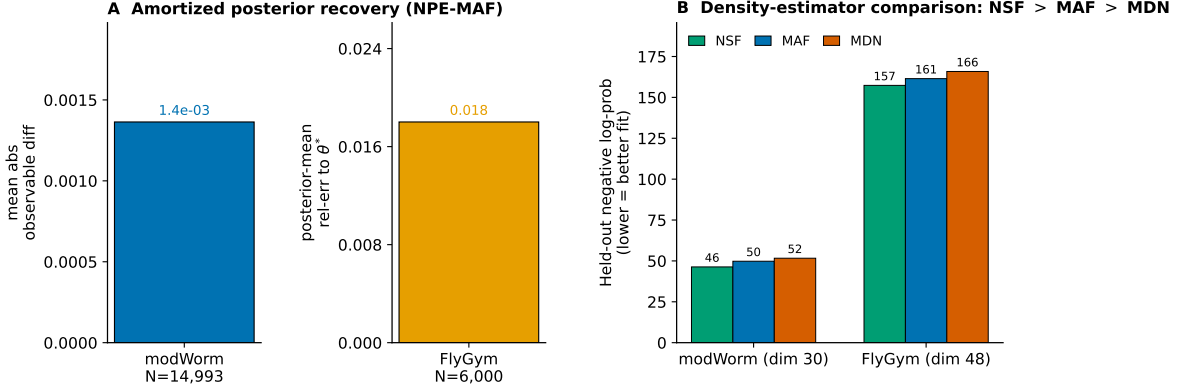


Figure 3: **The PGOB formulation with an amortized neural-posterior engine.** **a**, Posterior-predictive reproduction of each simulator’s own behaviour (modWorm mean absolute observable difference 1.4×10^{-3} ; FlyGym posterior-mean $\text{rel}_{\theta}=0.018$). **b**, Held-out negative log-probability for three density estimators (neural spline flow > MAF > mixture-density network on both simulators). The optimiser and the amortized posterior are interchangeable back-ends of one behaviour-only formulation.

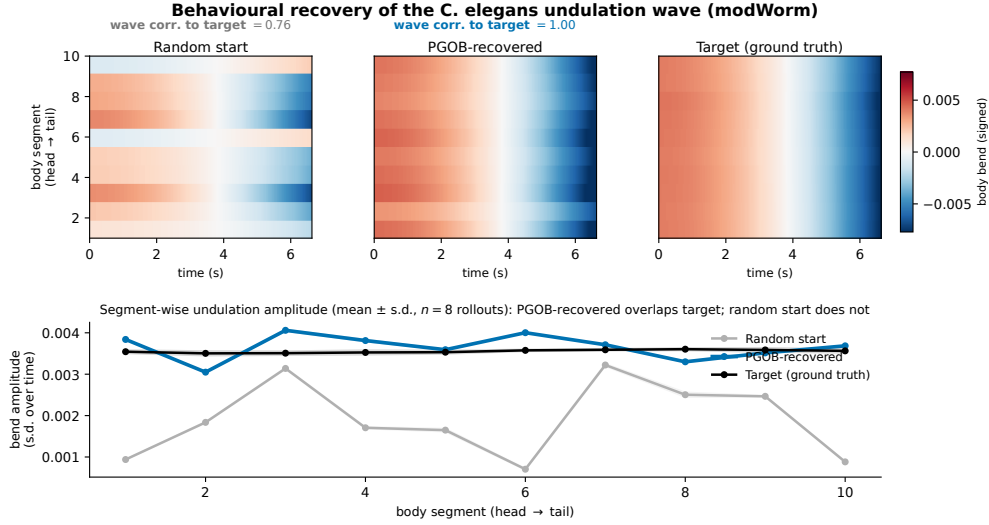


Figure 4: **Behaviour-level generation of the *C. elegans* undulation wave (modWorm).** Body-bend-oscillation kymographs (temporal mean removed) for a random scrambled start, the PGOB-generated parameters, and the simulator’s ground-truth target ($n=8$ seeds each); the generated worm reproduces the travelling wave (bend-wave correlation to target $0.76 \rightarrow 1.00$), the random start does not. The match is behavioural: the generated parameters lie on the behaviourally-equivalent set, not necessarily at the exact ground-truth coordinates.

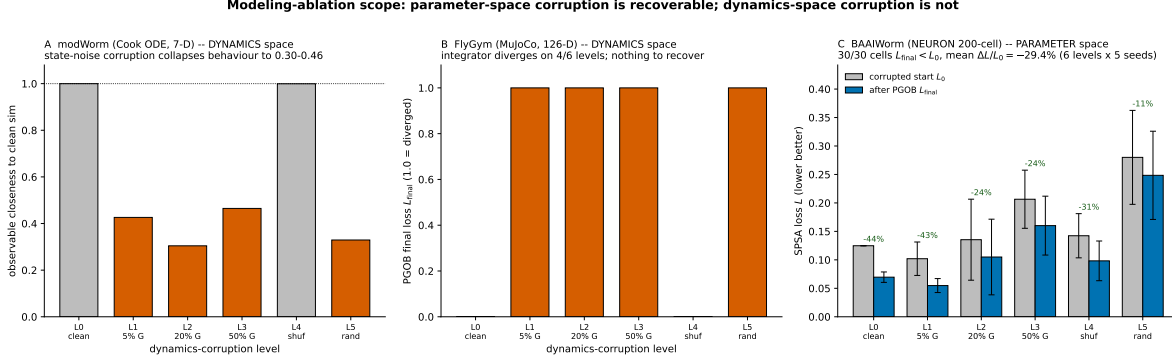


Figure 5: Scope of PGOB: parameter-space corruption is generated back, dynamics-space corruption is not. **a**, modWorm (Cook ODE) *dynamics*-space corruption: injecting state noise collapses observable closeness from 0.9999 to 0.30–0.46 (orange; grey are intact). **b**, FlyGym v2 (MuJoCo) *dynamics*-space corruption: the integrator diverges on 4/6 levels ($L_{final}=1.0$), leaving no usable forward model. **c**, BAAIWorm *parameter*-space corruption: corrupted-start loss L_0 vs. after-PGOB loss L_{final} per level; 30/30 cells improve, mean $\Delta L/L_0 = -29.4\%$ over the full L_0 – L_5 sweep. PGOB is a parameter-space generation framework, not a dynamics-space repair tool.

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